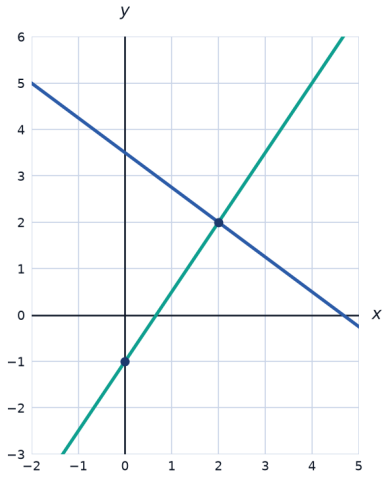
Radicals & Exponents

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Multiplying polynomials

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(5)$.

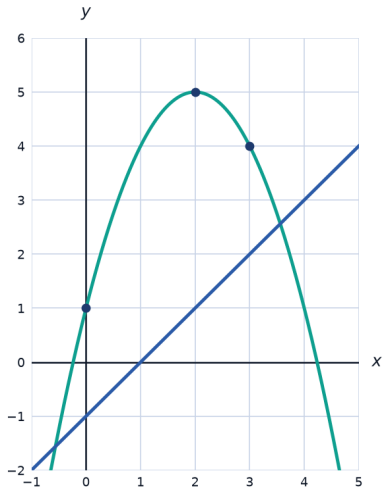


2 [Foundational] • Target: Multiplying polynomials

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

3 [Foundational] • Target: Special products

For the model $f(x) = 6x + 7$ in special products, evaluate $f(2)$.



4 [Foundational] • Target: Special products

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

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5 [Foundational] • Target: Special products

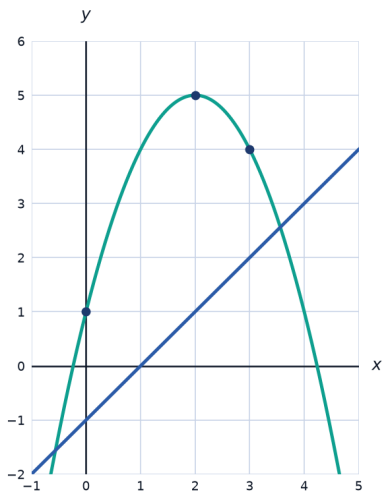
A data set recorded for special products consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Special products

For the model $f(x) = 6x + 7$ in special products, evaluate $f(3)$.

7 [Foundational] • Target: Special products

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Special products

For the model $f(x) = 6x + 7$ in special products, evaluate $f(4)$.

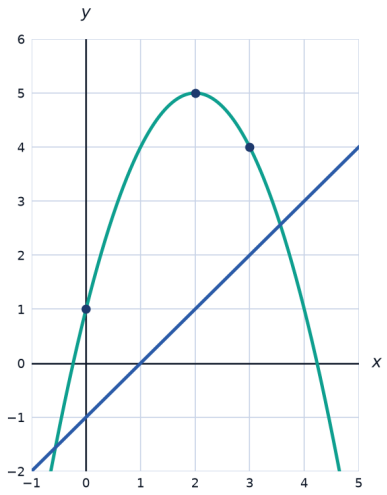
9 [Intermediate] • Target: Special products

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

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10 **[Intermediate]** • **Target: Special products**

For the model $f(x) = 6x + 7$ in special products, evaluate $f(5)$.

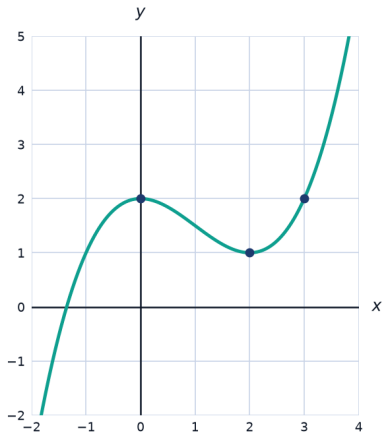


11 **[Intermediate]** • **Target: Special products**

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

12 **[Intermediate]** • **Target: Factoring out the greatest common factor (GCF)**

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(2)$.



13 **[Intermediate]** • **Target: Factoring out the greatest common factor (GCF)**

A data set recorded for factoring out the greatest common factor (gcf) consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

14 **[Intermediate]** • **Target: Factoring out the greatest common factor (GCF)**

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

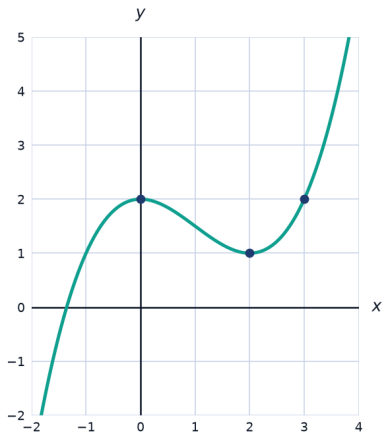
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15 [Challenging] • Target: Factoring out the greatest common factor (GCF) • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(4)$.

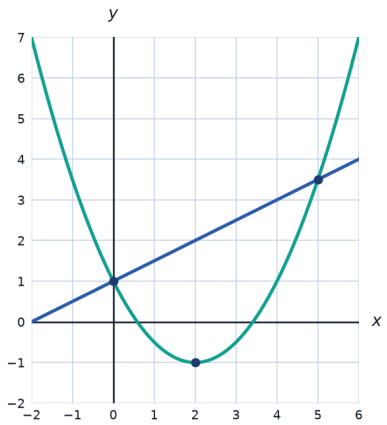
16 [Challenging] • Target: Factoring out the greatest common factor (GCF) • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(5)$.



17 [Challenging] • Target: Factoring trinomials • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the model $f(x) = 3x + 9$ in factoring trinomials, evaluate $f(2)$.



18 [Challenging] • Target: Factoring trinomials • Cross-Lesson: 6.01 Systems & Inequalities

A data set recorded for factoring trinomials consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

19 [Challenging] • Target: Factoring trinomials • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 3x + 9$ in factoring trinomials, evaluate $f(3)$.

20 [Challenging] • Target: Factoring trinomials • Cross-Lesson: 8.01 Factoring Polynomials

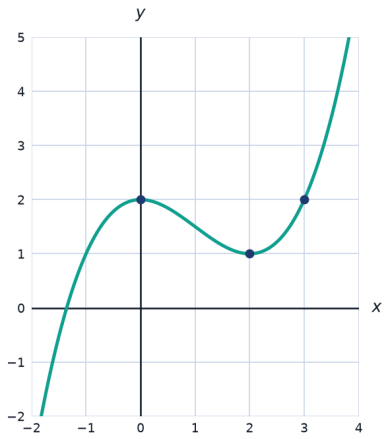
For the model $f(x) = 3x + 9$ in factoring trinomials, evaluate $f(4)$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Adding and subtracting polynomials

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(2)$.



a. 10

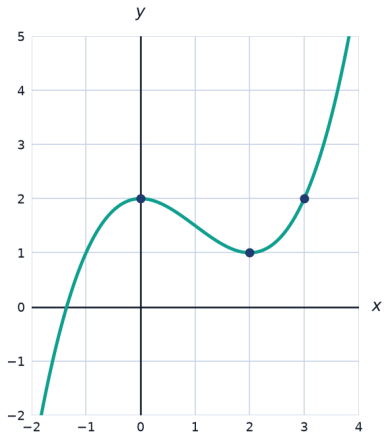
b. 13

c. 6

d. 14

22 [Foundational] • Target: Adding and subtracting polynomials

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

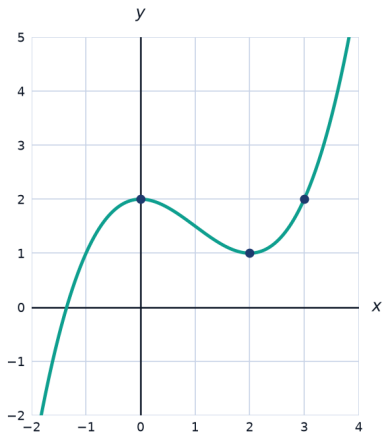
c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

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23 [Foundational] • Target: Adding and subtracting polynomials

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(5)$.



a. 19

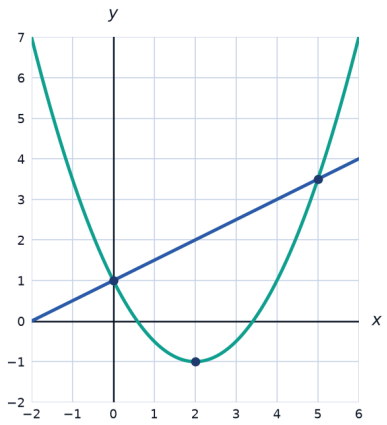
b. 22

c. 15

d. 23

24 [Foundational] • Target: Multiplying a polynomial by a monomial

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(2)$.



a. 13

b. 17

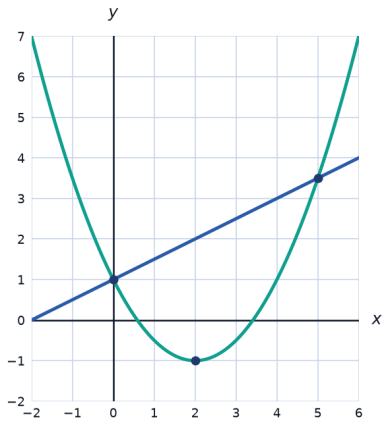
c. 8

d. 17

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25 [Foundational] • Target: Multiplying a polynomial by a monomial

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

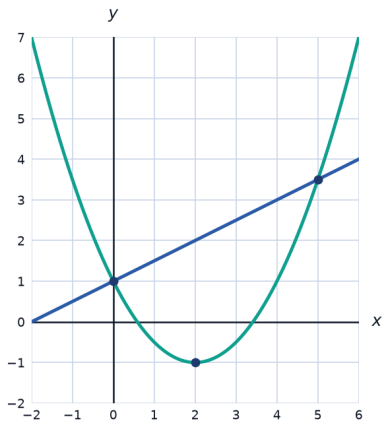
b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

26 [Foundational] • Target: Multiplying a polynomial by a monomial

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(5)$.



a. 25

b. 29

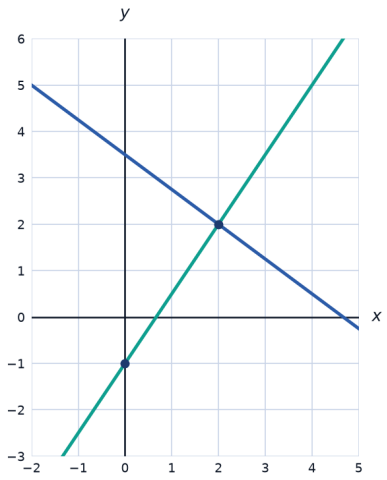
c. 20

d. 29

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27 [Foundational] • Target: Multiplying polynomials

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(2)$.



a. 16

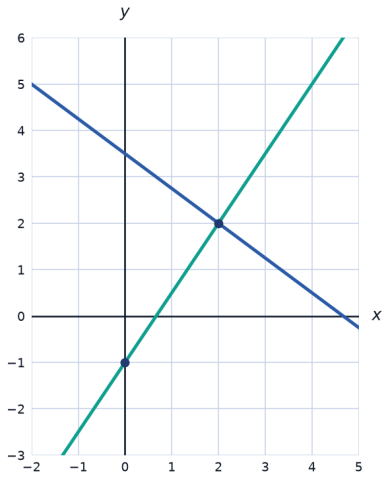
b. 21

c. 10

d. 20

28 [Foundational] • Target: Multiplying polynomials

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Adding and subtracting polynomials

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

30 [Intermediate] • Target: Adding and subtracting polynomials

A data set recorded for adding and subtracting polynomials consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

31 [Intermediate] • Target: Adding and subtracting polynomials

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

32 [Intermediate] • Target: Adding and subtracting polynomials

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(4)$.

a. 16

b. 19

c. 12

d. 20

33 [Intermediate] • Target: Adding and subtracting polynomials

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

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34 [Intermediate] • Target: Adding and subtracting polynomials

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

35 [Intermediate] • Target: Multiplying a polynomial by a monomial

A data set recorded for multiplying a polynomial by a monomial consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

36 [Intermediate] • Target: Multiplying a polynomial by a monomial

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(3)$.

a. 17

b. 21

c. 12

d. 21

37 [Intermediate] • Target: Multiplying a polynomial by a monomial

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(4)$.

a. 21

b. 25

c. 16

d. 25

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiplying a polynomial by a monomial • Cross-Lesson: 6.01 Systems & Inequalities

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

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39 [Challenging] • Target: Multiplying a polynomial by a monomial • Cross-Lesson: 7.01 Sequences & Series

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

40 [Challenging] • Target: Multiplying polynomials • Cross-Lesson: 8.01 Factoring Polynomials

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

41 [Challenging] • Target: Multiplying polynomials • Cross-Lesson: 1.01 Linear Equations & Lines

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(3)$.

a. 21

b. 26

c. 15

d. 25

42 [Challenging] • Target: Multiplying polynomials • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

43 [Challenging] • Target: Multiplying polynomials • Cross-Lesson: 11.01 Statistics & Data Analysis

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

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44 [Challenging] • Target: Special products • Cross-Lesson: 2.01 Quadratic Functions

A data set recorded for special products consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

45 [Challenging] • Target: Factoring out the greatest common factor (GCF) • Cross-Lesson: 3.01 Exponents & Polynomials

A data set recorded for factoring out the greatest common factor (gcf) consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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